Chapter 8. Fitting Regression Models

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1 Introduction

- Regression analysis studies relationships between a response y (dependent variable) and regressors $x_1, \ldots, x_k$ (independent variables); we build a model $y = f(x_1, \ldots, x_k) + \varepsilon$.
- Simple linear regression ($k = 1$): $y = \beta_0 + \beta_1 x + \varepsilon$.
- Polynomial regression ($k = 1$): $y = \beta_0 + \beta_1 x + \cdots + \beta_p x^p + \varepsilon$.
- Multiple linear regression ($k \geq 2$): $y = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + \varepsilon$.

Steps for a simple linear regression analysis

1. Plot a scatter diagram (y vs x). If linear go to step 2; if not, consider a nonlinear $f(x)$ or transform to linearize.
2. Estimate β_0, β_1.
3. Perform residual analysis to assess model adequacy.
4. Go back to step 1 if the model does not pass the residual analysis.

2 Linear Regression Models

2.1 Simple linear regression

$y = \beta_0 + \beta_1 x + \varepsilon$, errors iid $(0, \sigma^2)$. **Important**: x is *not* random; y is. Hence

$$\mathrm{E}(y \mid x) = \beta_0 + \beta_1 x, \qquad \mathrm{Var}(y \mid x) = \sigma^2;$$

the true regression line is the line of mean responses; β_0 (intercept) and β_1 (slope) are the regression coefficients. Example: viscosity (y) of a polymer vs temperature (x).

Theorem 8.1: Scalar least squares

Minimizing $L = \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i)^2$ gives

$$\hat\beta_1 = \frac{S_{xy}}{S_{xx}} = \frac{\sum (x_i - \bar x)(y_i - \bar y)}{\sum (x_i - \bar x)^2}, \qquad \hat\beta_0 = \bar y - \hat\beta_1 \bar x.$$

Proof [Supplement]. $\frac{\partial L}{\partial \beta_0} = -2\sum (y_i - \hat\beta_0 - \hat\beta_1 x_i) = 0 \Rightarrow n\hat\beta_0 + \hat\beta_1 \sum x_i = \sum y_i \Rightarrow \hat\beta_0 = \bar y - \hat\beta_1 \bar x$. $\frac{\partial L}{\partial \beta_1} = -2\sum x_i (y_i - \hat\beta_0 - \hat\beta_1 x_i) = 0$. Substitute $\hat\beta_0$: $\sum x_i [(y_i - \bar y) - \hat\beta_1 (x_i - \bar x)] = 0$. Since $\sum \bar x (y_i - \bar y) = 0$ and $\sum \bar x (x_i - \bar x) = 0$, this is $\sum (x_i - \bar x)(y_i - \bar y) = \hat\beta_1 \sum (x_i - \bar x)^2$. ■

2.2 Multiple linear regression in matrix form

Data $(y_i, x_{i1}, \ldots, x_{ik})$, $i = 1, \ldots, n$: $y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik} + \varepsilon_i$, i.e.

$$\mathbf{y} = \mathbf{X}\boldsymbol\beta + \boldsymbol\varepsilon, \quad \mathbf{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}, \ \mathbf{X} = \begin{pmatrix} 1 & x_{11} & \cdots & x_{1k} \\ \vdots & & & \vdots \\ 1 & x_{n1} & \cdots & x_{nk} \end{pmatrix}, \ \boldsymbol\beta = \begin{pmatrix} \beta_0 \\ \vdots \\ \beta_k \end{pmatrix}, \ \boldsymbol\varepsilon \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n),$$

$\mathbf{y}$: $n \times 1$, $\mathbf{X}$: $n \times p$ $(p = k + 1)$, $\boldsymbol\beta$: $p \times 1$.

3 Least Squares Estimation

Motivation. Every design in the course is a special case of the linear regression model. Least squares gives the estimates and their covariance in one formula, which unifies the separate ANOVA computations.

Theorem 8.2: Normal equations

$$L = \sum_{i=1}^{n} \varepsilon_i^2 = \boldsymbol\varepsilon^T \boldsymbol\varepsilon = (\mathbf{y} - \mathbf{X}\boldsymbol\beta)^T (\mathbf{y} - \mathbf{X}\boldsymbol\beta) \quad \Longrightarrow \quad \mathbf{X}^T\mathbf{X}\hat{\boldsymbol\beta} = \mathbf{X}^T\mathbf{y}, \quad \hat{\boldsymbol\beta} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}.$$

Proof [Supplement].

$$L = \mathbf{y}^T\mathbf{y} - \boldsymbol\beta^T\mathbf{X}^T\mathbf{y} - \mathbf{y}^T\mathbf{X}\boldsymbol\beta + \boldsymbol\beta^T\mathbf{X}^T\mathbf{X}\boldsymbol\beta = \mathbf{y}^T\mathbf{y} - 2\boldsymbol\beta^T\mathbf{X}^T\mathbf{y} + \boldsymbol\beta^T\mathbf{X}^T\mathbf{X}\boldsymbol\beta,$$

$$\frac{\partial L}{\partial \boldsymbol\beta}\Big|_{\hat{\boldsymbol\beta}} = -2\mathbf{X}^T\mathbf{y} + 2\mathbf{X}^T\mathbf{X}\hat{\boldsymbol\beta} = \mathbf{0}.$$

(Using $\partial(\mathbf{a}^T\boldsymbol\beta)/\partial\boldsymbol\beta = \mathbf{a}$ and $\partial(\boldsymbol\beta^T\mathbf{A}\boldsymbol\beta)/\partial\boldsymbol\beta = 2\mathbf{A}\boldsymbol\beta$ for symmetric $\mathbf{A}$.) The Hessian $2\mathbf{X}^T\mathbf{X}$ is positive definite when $\mathbf{X}$ has full column rank, so this is a minimum. ■

Fitted values $\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y} = \mathbf{H}\mathbf{y}$ (hat matrix $\mathbf{H}$, $n \times n$). Residuals $\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} = (\mathbf{I} - \mathbf{H})\mathbf{y}$.

Theorem 8.3: Properties of $\hat{\beta}$

If $\mathrm{E}(\boldsymbol{\varepsilon}) = \mathbf{0}$ and $\mathrm{Var}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}_n$,

$$\mathrm{E}(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}, \qquad \mathrm{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}^T\mathbf{X})^{-1}.$$

Proof [Supplement]. $\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) = \boldsymbol{\beta} + (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\boldsymbol{\varepsilon}$. Hence $\mathrm{E}(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}$ and

$$\mathrm{Var}(\hat{\boldsymbol{\beta}}) = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\,\sigma^2\mathbf{I}\,\mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1} = \sigma^2(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1} = \sigma^2(\mathbf{X}^T\mathbf{X})^{-1}.$$

■

3.1 Estimating σ^2

Theorem 8.4

$$SS_E = \sum_{i=1}^{n} e_i^2 = \mathbf{e}^T\mathbf{e} = \mathbf{y}^T\mathbf{y} - \hat{\boldsymbol{\beta}}^T\mathbf{X}^T\mathbf{y}, \qquad \hat{\sigma}^2 = MS_E = \frac{SS_E}{n-k-1}, \qquad \mathrm{E}(MS_E) = \sigma^2.$$

Proof [Supplement].

$$\begin{aligned} SS_E &= (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})^T(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \mathbf{y}^T\mathbf{y} - 2\hat{\boldsymbol{\beta}}^T\mathbf{X}^T\mathbf{y} + \hat{\boldsymbol{\beta}}^T\mathbf{X}^T\mathbf{X}\hat{\boldsymbol{\beta}} \\ &= \mathbf{y}^T\mathbf{y} - 2\hat{\boldsymbol{\beta}}^T\mathbf{X}^T\mathbf{y} + \hat{\boldsymbol{\beta}}^T\mathbf{X}^T\mathbf{y} \qquad (\mathbf{X}^T\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}^T\mathbf{y}) \\ &= \mathbf{y}^T\mathbf{y} - \hat{\boldsymbol{\beta}}^T\mathbf{X}^T\mathbf{y}. \end{aligned}$$

Also $SS_E = \mathbf{y}^T(\mathbf{I} - \mathbf{H})\mathbf{y}$, so by $\mathrm{E}(\mathbf{y}^T\mathbf{A}\mathbf{y}) = \mathrm{tr}(\mathbf{A}\mathbf{V}) + \boldsymbol{\mu}^T\mathbf{A}\boldsymbol{\mu}$,

$$\mathrm{E}(SS_E) = \sigma^2\mathrm{tr}(\mathbf{I} - \mathbf{H}) + \boldsymbol{\beta}^T\mathbf{X}^T(\mathbf{I} - \mathbf{H})\mathbf{X}\boldsymbol{\beta} = \sigma^2(n-p) + 0 = \sigma^2(n-k-1).$$

■

Example 8.5: A two-regressor fit

With $n = 16$ observations and two regressors,

$$\mathbf{X}^T\mathbf{X} = \begin{pmatrix} 16 & 1458 & 164 \\ 1458 & 133560 & 14946 \\ 164 & 14946 & 1726 \end{pmatrix}, \qquad \mathbf{X}^T\mathbf{y} = \begin{pmatrix} 37577 \\ 3429550 \\ 385562 \end{pmatrix},$$

solving $\mathbf{X}^T\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}^T\mathbf{y}$ gives $\hat{\boldsymbol{\beta}} = (1566.078,\ 7.621,\ 8.585)^T$, i.e. $\hat{y} = 1566.08 + 7.62x_1 + 8.58x_2$ (same as `lm(y~x1+x2)`).

4 Regression Approach to the CRD with a Single Factor

Motivation. Fitting the one-way ANOVA as a regression with indicator variables reproduces the treatment sum of squares, which shows that the two approaches are the same model.

4.1 Effects coding (constraint $\sum \tau_i = 0$)

$y_{ij} = \mu + \tau_i + \varepsilon_{ij}$, $\sum_{i=1}^{a} \tau_i = 0$. For $a = 3$, $n = 2$:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \mathbf{y} = \begin{pmatrix} y_{11} \\ y_{12} \\ y_{21} \\ y_{22} \\ y_{31} \\ y_{32} \end{pmatrix}, \quad \mathbf{X} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \\ 1 & -1 & -1 \\ 1 & -1 & -1 \end{pmatrix}, \quad \boldsymbol{\beta} = \begin{pmatrix} \mu \\ \tau_1 \\ \tau_2 \end{pmatrix},$$

i.e. $y = \mu + \tau_1 x_1 + \tau_2 x_2 + \varepsilon$ with $x_1 = 1$ (level 1), -1 (level 3), 0 otherwise; $x_2 = 1$ (level 2), -1 (level 3), 0 otherwise.

Theorem 8.6: Effects coding reproduces the ANOVA estimates

For general a (balanced, n per level) with $x_i = 1$ at level i, -1 at level a, 0 otherwise ($i = 1, \ldots, a-1$):

$$\hat{\mu} = \bar{y}_{\cdot\cdot}, \qquad \hat{\tau}_i = \bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot} \quad (i = 1, \ldots, a-1), \qquad \hat{\tau}_a = -\sum_{i<a} \hat{\tau}_i = \bar{y}_{a\cdot} - \bar{y}_{\cdot\cdot}.$$

Proof [Supplement]. Entries of $\mathbf{X}^T\mathbf{X}$: $(\mathbf{1}, \mathbf{1}) = N$; $(\mathbf{1}, x_i) = n - n = 0$; $(x_i, x_i) = n + n = 2n$; $(x_i, x_l) = n$ for $i \neq l$ (both equal -1 on the n runs at level a). So

$$\mathbf{X}^T\mathbf{X} = \begin{pmatrix} N & \mathbf{0}^T \\ \mathbf{0} & n(\mathbf{I}_{a-1} + \mathbf{J}_{a-1}) \end{pmatrix}, \qquad \mathbf{X}^T\mathbf{y} = \begin{pmatrix} y_{\cdot\cdot} \\ y_{1\cdot} - y_{a\cdot} \\ \vdots \\ y_{a-1,\cdot} - y_{a\cdot} \end{pmatrix}.$$

Since $(\mathbf{I} + \mathbf{J})(\mathbf{I} - \frac{1}{a}\mathbf{J}) = \mathbf{I} + \mathbf{J} - \frac{1}{a}\mathbf{J} - \frac{a-1}{a}\mathbf{J} = \mathbf{I}$ (as $\mathbf{J}\mathbf{J} = (a-1)\mathbf{J}$), $[n(\mathbf{I} + \mathbf{J})]^{-1} = \frac{1}{n}(\mathbf{I} - \frac{1}{a}\mathbf{J})$. Thus $\hat{\mu} = y_{\cdot\cdot}/N$ and

$$\hat{\tau}_i = \frac{1}{n}\Big[(y_{i\cdot} - y_{a\cdot}) - \frac{1}{a}\sum_{l=1}^{a-1}(y_{l\cdot} - y_{a\cdot})\Big] = \frac{1}{n}\Big[y_{i\cdot} - y_{a\cdot} - \frac{y_{\cdot\cdot} - a\,y_{a\cdot}}{a}\Big] = \frac{y_{i\cdot}}{n} - \frac{y_{\cdot\cdot}}{an} = \bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}.$$

■

Example 8.7: Four treatments, $n = 4$

Trt	Observations				$\bar{y}_{i\cdot}$
1	3129	3000	2865	2890	2971.00
2	3200	3300	2975	3150	3156.25
3	2800	2900	2985	3050	2933.75
4	2600	2700	2600	2765	2666.25

$y = \mu + \tau_1 x_1 + \tau_2 x_2 + \tau_3 x_3 + \varepsilon$ with effects coding ($x_i = -1$ for level 4). The 16×4 design matrix has rows $(1,1,0,0) \times 4$, $(1,0,1,0) \times 4$, $(1,0,0,1) \times 4$, $(1,-1,-1,-1) \times 4$ and

$$\mathbf{X}^T\mathbf{X} = \begin{pmatrix} 16 & 0 & 0 & 0 \\ 0 & 8 & 4 & 4 \\ 0 & 4 & 8 & 4 \\ 0 & 4 & 4 & 8 \end{pmatrix}, \qquad \mathbf{X}^T\mathbf{y} = \begin{pmatrix} 46909 \\ 1219 \\ 1960 \\ 1070 \end{pmatrix}$$

($1219 = y_{1\cdot} - y_{4\cdot} = 11884 - 10665$, etc.). Solving: $\hat{\mu} = 2931.8125$, $\hat{\tau}_1 = 39.1875$, $\hat{\tau}_2 = 224.4375$, $\hat{\tau}_3 = 1.9375$, and $\hat{\tau}_4 = -(39.1875 + 224.4375 + 1.9375) = -265.5625$ — exactly $\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}$.

4.2 Reference-cell (dummy) coding (constraint $\tau_1 = 0$)

$$\mathbf{X} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix}, \qquad \boldsymbol{\beta} = \begin{pmatrix} \mu \\ \tau_2 \\ \tau_3 \end{pmatrix}, \qquad x_1 = \mathbb{1}(\text{level 2}),\ x_2 = \mathbb{1}(\text{level 3}).$$

Now $\hat{\mu} = \bar{y}_{1\cdot}$ and $\hat{\tau}_i = \bar{y}_{i\cdot} - \bar{y}_{1\cdot}$: μ is the mean of the reference level. The fitted values $\hat{y}_{ij} = \bar{y}_{i\cdot}$, the residuals and all SS are identical under both codings (same column space), only the meaning of the parameters differs.

Remark 8.8: fitted values do not depend on the coding

CRD with $a = 4$, $n = 2$, $\mathbf{y} = (5, 5, 0, 0, 0, 0, -5, -5)^T$ and effects coding. Since $\hat{y}_{ij} = \bar{y}_{i\cdot}$ for any full-rank coding of the one-way model: $\bar{y}_{1\cdot} = 5$, $\bar{y}_{2\cdot} = 0$, $\bar{y}_{3\cdot} = 0$, $\bar{y}_{4\cdot} = -5$, so $\hat{\mathbf{y}} = (5, 5, 0, 0, 0, 0, -5, -5)^T$ and $\mathbf{e} = \mathbf{0}$. (Directly: $\hat{\mu} = 0$, $\hat{\tau} = (5, 0, 0)$, $\hat{\tau}_4 = -5$.)

5 Regression Approach to the 2^2 Design

$$y_{ijk} = \mu + A_i + B_j + (AB)_{ij} + \varepsilon_{ijk} \ \longleftrightarrow\ y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2 + \varepsilon,$$

with the chemical-process cell averages:

$$\mathbf{y} = \begin{pmatrix} 80/3 \\ 100/3 \\ 60/3 \\ 90/3 \end{pmatrix}, \quad \mathbf{X} = \begin{pmatrix} 1 & -1 & -1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & 1 & 1 \end{pmatrix}, \quad \mathbf{X}^T\mathbf{X} = 4\mathbf{I}_4,$$

$\hat{\boldsymbol{\beta}} = \frac{1}{4}\mathbf{X}^T\mathbf{y} = (27.5,\ 4.1667,\ -2.5,\ 0.8333)^T$: the coefficients are half the effects $A = 8.33$, $B = -5.00$, $AB = 1.67$ (Ch. 6).

6 Test for Significance of Regression

Motivation. The overall F-test of regression asks whether any regressor matters, the analogue of testing all treatment means equal.

$H_0 : \beta_1 = \beta_2 = \cdots = \beta_k = 0$ vs $H_1 : \beta_j \neq 0$ for at least one j. Rejecting H_0 means at least one regressor contributes significantly.

Theorem 8.9: ANOVA decomposition

$$\underbrace{\mathbf{y}^T\mathbf{y} - \frac{(\sum y_i)^2}{n}}_{SS_T} = \underbrace{\hat{\boldsymbol{\beta}}^T\mathbf{X}^T\mathbf{y} - \frac{(\sum y_i)^2}{n}}_{SS_R} + \underbrace{\mathbf{y}^T\mathbf{y} - \hat{\boldsymbol{\beta}}^T\mathbf{X}^T\mathbf{y}}_{SS_E}, \qquad (n-1) = k + (n-k-1).$$

Proof [Supplement]. Add and subtract $\hat{\boldsymbol{\beta}}^T\mathbf{X}^T\mathbf{y}$. In matrix form $SS_T = \mathbf{y}^T(\mathbf{I} - \frac{1}{n}\mathbf{J})\mathbf{y}$, $SS_R = \mathbf{y}^T(\mathbf{H} - \frac{1}{n}\mathbf{J})\mathbf{y}$, $SS_E = \mathbf{y}^T(\mathbf{I} - \mathbf{H})\mathbf{y}$ (because $\hat{\boldsymbol{\beta}}^T\mathbf{X}^T\mathbf{y} = \mathbf{y}^T\mathbf{H}\mathbf{y}$ and $\mathbf{1} \in \Omega(\mathbf{X})$). $\mathbf{H} - \frac{1}{n}\mathbf{J}$ and $\mathbf{I} - \mathbf{H}$ are idempotent with ranks k and $n-k-1$ and $(\mathbf{I} - \mathbf{H})(\mathbf{H} - \frac{1}{n}\mathbf{J}) = \mathbf{0}$, so (Theorems 3.11 and 3.12) under H_0 (where $(\mathbf{H} - \frac{1}{n}\mathbf{J})\mathbf{X}\boldsymbol{\beta} = \beta_0\mathbf{1} - \beta_0\mathbf{1} = \mathbf{0}$), $SS_R/\sigma^2 \sim \chi^2_k$, $SS_E/\sigma^2 \sim \chi^2_{n-k-1}$, independent. ■

$$MS_R = \frac{SS_R}{k}, \qquad MS_E = \frac{SS_E}{n-k-1}, \qquad F_0 = \frac{MS_R}{MS_E} \overset{H_0}{\sim} F_{k,n-k-1}; \quad \text{reject if } F_0 > F_{\alpha,k,n-k-1}.$$

Source	SS	DF	MS	F_0
Regression	$\hat{\boldsymbol{\beta}}^T\mathbf{X}^T\mathbf{y} - \frac{(\sum y_i)^2}{n}$	k	MS_R	MS_R/MS_E
Error	$\mathbf{y}^T\mathbf{y} - \hat{\boldsymbol{\beta}}^T\mathbf{X}^T\mathbf{y}$	$n-k-1$	MS_E	
Total	$\mathbf{y}^T\mathbf{y} - \frac{(\sum y_i)^2}{n}$	$n-1$		

6.1 Tests on individual coefficients

$H_0 : \beta_j = 0$ vs $H_1 : \beta_j \neq 0$:

$$t_0 = \frac{\hat{\beta}_j}{\sqrt{\hat{\sigma}^2 C_{jj}}} = \frac{\hat{\beta}_j}{se(\hat{\beta}_j)}, \qquad C_{jj} = \left[(\mathbf{X}^T\mathbf{X})^{-1}\right]_{jj};$$

reject if $|t_0| > t_{\alpha/2,n-k-1}$.

Why t [Supplement]. $\hat{\beta}_j \sim N(\beta_j, \sigma^2 C_{jj})$; $(n-p)\hat{\sigma}^2/\sigma^2 \sim \chi^2_{n-p}$; and $\hat{\boldsymbol{\beta}} \perp\!\!\!\perp SS_E$ because

$$\mathrm{Cov}(\hat{\boldsymbol{\beta}}, \mathbf{e}) = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\sigma^2\mathbf{I}(\mathbf{I} - \mathbf{H}) = \sigma^2(\mathbf{X}^T\mathbf{X})^{-1}(\mathbf{X}^T - \mathbf{X}^T\mathbf{H}) = \mathbf{0}$$

($\mathbf{X}^T\mathbf{H} = (\mathbf{H}\mathbf{X})^T = \mathbf{X}^T$). Hence $\dfrac{(\hat{\beta}_j - \beta_j)/\sqrt{\sigma^2 C_{jj}}}{\sqrt{\hat{\sigma}^2/\sigma^2}} = \dfrac{\hat{\beta}_j - \beta_j}{\sqrt{\hat{\sigma}^2 C_{jj}}} \sim t_{n-p}$. ■

7 Confidence Intervals in Regression

On coefficients. $\dfrac{\hat{\beta}_j - \beta_j}{\sqrt{\hat{\sigma}^2 C_{jj}}} \sim t_{n-(k+1)}$, $j = 0, \ldots, k$, so

$$\hat{\beta}_j - t_{\alpha/2,n-(k+1)}se(\hat{\beta}_j) \le \beta_j \le \hat{\beta}_j + t_{\alpha/2,n-(k+1)}se(\hat{\beta}_j).$$

On the mean response at $\mathbf{x}_0 = (1, x_{01}, \ldots, x_{0k})^T$:

1. $\hat{y}_0 = \mathbf{x}_0^T\hat{\boldsymbol{\beta}}$;
2. $\mathrm{E}(\hat{y}_0) = \mathbf{x}_0^T\boldsymbol{\beta} = \mathrm{E}(y \mid \mathbf{x}_0)$, $\mathrm{Var}(\hat{y}_0) = \mathbf{x}_0^T\mathrm{Var}(\hat{\boldsymbol{\beta}})\mathbf{x}_0 = \sigma^2\mathbf{x}_0^T(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{x}_0$;
3. $\dfrac{\hat{y}_0 - \mathrm{E}(y \mid \mathbf{x}_0)}{\sqrt{\hat{\sigma}^2\mathbf{x}_0^T(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{x}_0}} \sim t_{n-(k+1)}$;
4. $100(1-\alpha)\%$ CI:

$$\hat{y}_0 - t_{\alpha/2,n-(k+1)}\sqrt{\hat{\sigma}^2\mathbf{x}_0^T(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{x}_0} \le \mathrm{E}(y \mid \mathbf{x}_0) \le \hat{y}_0 + t_{\alpha/2,n-(k+1)}\sqrt{\hat{\sigma}^2\mathbf{x}_0^T(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{x}_0}.$$

8 The Extra Sum of Squares Principle (General)

Partition $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}$ (full model) and test $H_0 : \boldsymbol{\beta}_1 = \mathbf{0}$ via the reduced model $\mathbf{y} = \mathbf{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}$:

$$F_0 = \frac{[SS_E(\mathrm{R}) - SS_E(\mathrm{F})]/[df(\mathrm{R}) - df(\mathrm{F})]}{SS_E(\mathrm{F})/df(\mathrm{F})} \overset{H_0}{\sim} F_{df(\mathrm{R})-df(\mathrm{F}),\ df(\mathrm{F})}.$$

Proof: $SS_E(\mathrm{R}) - SS_E(\mathrm{F}) = \mathbf{y}^T(\mathbf{H}_F - \mathbf{H}_R)\mathbf{y}$, with $\mathbf{H}_F - \mathbf{H}_R$ idempotent, orthogonal to $\mathbf{I} - \mathbf{H}_F$ and $(\mathbf{H}_F - \mathbf{H}_R)\mathbf{X}\boldsymbol{\beta} = \mathbf{0}$ under H_0 (exactly as for the RCBD in Ch. 4). This principle underlies the RCBD, BIBD (adjusted SS), Tukey's 1-df test and ANCOVA.

9 Selected exercises

Motivation. Regression is also a design tool: where the x's are placed determines the precision of the estimates. The exercises find the best placement for a slope, extend least squares to unequal variances, and give the leverage identities behind residual diagnostics.

Exercise 8.1 (Where to put the points; Montgomery 10.15)

A straight line $y = \beta_0 + \beta_1 x + \varepsilon$ is fitted with n (even) runs at levels $x_1, \ldots, x_n$ in the coded region $[-1, 1]$. Show that $\mathrm{Var}(\hat{\beta}_1)$ is minimized by putting $n/2$ runs at each end, and discuss the price of this design.

Solution.

Step 1. Since $\sum_i (x_i - \bar{x})\bar{y} = 0$, the scalar least squares estimate is $\hat{\beta}_1 = \sum_i (x_i - \bar{x})y_i / S_{xx}$, a linear combination of independent y_i, so $\mathrm{Var}(\hat{\beta}_1) = \sigma^2 \sum_i (x_i - \bar{x})^2 / S_{xx}^2 = \sigma^2 / S_{xx}$ (with $S_{xx} > 0$, i.e. not all x_i equal). We therefore maximize S_{xx}.

Step 2. $S_{xx} = \sum_i x_i^2 - n\bar{x}^2 \le \sum_i x_i^2 \le n$, since $|x_i| \le 1$. Equality in both requires $x_i^2 = 1$ for all i and $\bar{x} = 0$, i.e. half the runs at -1 and half at $+1$; this design attains $S_{xx} = n$ and $\mathrm{Var}(\hat{\beta}_1) = \sigma^2/n$.

Step 3 (price). With only two distinct levels, a quadratic $\beta_{11}x^2$ is indistinguishable from a shift of the intercept, so lack of fit cannot be detected. Adding center runs (as in Ch. 6) sacrifices a little precision to check curvature; the end-point design is advisable only when linearity is certain. ■

Exercise 8.2 (Weighted least squares; Montgomery 10.16)

Let $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $\mathbf{X}$ of full rank p and $\mathrm{Var}(\varepsilon_i) = \sigma^2/w_i$ (independent; known $w_i > 0$), and $\mathbf{W} = \mathrm{diag}(w_1, \ldots, w_n)$.
(a) Show that minimizing $\sum_i w_i(y_i - \mathbf{x}_i^T\boldsymbol{\beta})^2$ gives $\hat{\boldsymbol{\beta}}_W = (\mathbf{X}^T\mathbf{W}\mathbf{X})^{-1}\mathbf{X}^T\mathbf{W}\mathbf{y}$. For a straight line, write the two normal equations.
(b) Show $\mathrm{E}\hat{\boldsymbol{\beta}}_W = \boldsymbol{\beta}$ and $\mathrm{Var}(\hat{\boldsymbol{\beta}}_W) = \sigma^2(\mathbf{X}^T\mathbf{W}\mathbf{X})^{-1}$.

Solution.

(a)

Step 1. With $\mathbf{W}^{1/2} = \mathrm{diag}(\sqrt{w_i})$, $\mathbf{y}^* = \mathbf{W}^{1/2}\mathbf{y}$ and $\mathbf{X}^* = \mathbf{W}^{1/2}\mathbf{X}$, the criterion is $\|\mathbf{y}^* - \mathbf{X}^*\boldsymbol{\beta}\|^2$, an ordinary least squares problem; $\mathbf{X}^*$ has full rank because $\mathbf{W}^{1/2}$ is invertible.

Step 2. By the normal-equations theorem, $\mathbf{X}^{*T}\mathbf{X}^*\hat{\boldsymbol{\beta}} = \mathbf{X}^{*T}\mathbf{y}^*$, i.e. $\mathbf{X}^T\mathbf{W}\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}^T\mathbf{W}\mathbf{y}$, with the stated unique solution. For $y = \beta_0 + \beta_1 x$,

$$\hat{\beta}_0 \sum_i w_i + \hat{\beta}_1 \sum_i w_i x_i = \sum_i w_i y_i, \qquad \hat{\beta}_0 \sum_i w_i x_i + \hat{\beta}_1 \sum_i w_i x_i^2 = \sum_i w_i x_i y_i.$$

(b) $\hat{\boldsymbol{\beta}}_W = \boldsymbol{\beta} + (\mathbf{X}^T\mathbf{W}\mathbf{X})^{-1}\mathbf{X}^T\mathbf{W}\boldsymbol{\varepsilon}$, so $\mathrm{E}\hat{\boldsymbol{\beta}}_W = \boldsymbol{\beta}$, and with $\mathrm{Var}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{W}^{-1}$,

$$\mathrm{Var}(\hat{\boldsymbol{\beta}}_W) = (\mathbf{X}^T\mathbf{W}\mathbf{X})^{-1}\mathbf{X}^T\mathbf{W}(\sigma^2\mathbf{W}^{-1})\mathbf{W}\mathbf{X}(\mathbf{X}^T\mathbf{W}\mathbf{X})^{-1} = \sigma^2(\mathbf{X}^T\mathbf{W}\mathbf{X})^{-1}.$$

The transformed errors $\mathbf{W}^{1/2}\boldsymbol{\varepsilon}$ have covariance $\sigma^2\mathbf{I}$, so all the ordinary least squares results (tests, intervals) apply to the starred model; observations with small variance get large weight. ■

Exercise 8.3 (Leverage and deleted residuals)

In the model with an intercept, let h_{ii} be the diagonal entries of $\mathbf{H} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$ and e_i the residuals.
(a) Show $\sum_i h_{ii} = p$, $\frac{1}{n} \le h_{ii} \le 1$ and $\mathrm{Var}(e_i) = \sigma^2(1 - h_{ii})$.
(b) Let $\hat{y}_{(i)}$ be the prediction of y_i from the fit without observation i (assume $h_{ii} < 1$). Show $y_i - \hat{y}_{(i)} = e_i/(1 - h_{ii})$, so PRESS $= \sum_i \{e_i/(1 - h_{ii})\}^2$ needs only one fit.

Solution [Supplement].

(a)

Step 1. $\mathbf{H}$ is symmetric and idempotent, and $\mathrm{tr}\mathbf{H} = \mathrm{tr}\{(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{X}\} = \mathrm{tr}\mathbf{I}_p = p$.

Step 2. $h_{ii} = (\mathbf{H}^2)_{ii} = \sum_j h_{ij}^2 \ge h_{ii}^2$, so $0 \le h_{ii} \le 1$. Because $\mathbf{1} \in$ the column space of $\mathbf{X}$, $\mathbf{H} - \frac{1}{n}\mathbf{J}$ is also symmetric idempotent ($\mathbf{HJ} = \mathbf{JH} = \mathbf{J}$), so its diagonal entries $h_{ii} - \frac{1}{n}$ are ≥ 0 by the same argument.

Step 3. $\mathbf{e} = (\mathbf{I} - \mathbf{H})\mathbf{y}$ and $\mathrm{Var}(\mathbf{e}) = \sigma^2(\mathbf{I} - \mathbf{H})(\mathbf{I} - \mathbf{H})^T = \sigma^2(\mathbf{I} - \mathbf{H})$, whose diagonal is $\sigma^2(1 - h_{ii})$. High-leverage points have small residual variance, which is why residuals are standardized by $\sqrt{MS_E(1 - h_{ii})}$.

(b)

Step 1. Let $\mathbf{A} = \mathbf{X}^T\mathbf{X}$ and $\mathbf{x}_i^T$ the ith row. Without observation i, $\mathbf{X}_{(i)}^T\mathbf{X}_{(i)} = \mathbf{A} - \mathbf{x}_i\mathbf{x}_i^T$ and $\mathbf{X}_{(i)}^T\mathbf{y}_{(i)} = \mathbf{X}^T\mathbf{y} - \mathbf{x}_i y_i$. Multiplying out shows (Sherman–Morrison)

$$(\mathbf{A} - \mathbf{x}_i\mathbf{x}_i^T)^{-1} = \mathbf{A}^{-1} + \frac{\mathbf{A}^{-1}\mathbf{x}_i\mathbf{x}_i^T\mathbf{A}^{-1}}{1 - h_{ii}}, \qquad h_{ii} = \mathbf{x}_i^T\mathbf{A}^{-1}\mathbf{x}_i.$$

Step 2. Hence, with $\hat{y}_i = \mathbf{x}_i^T\mathbf{A}^{-1}\mathbf{X}^T\mathbf{y}$,

$$\begin{aligned}\hat{y}_{(i)} &= \mathbf{x}_i^T(\mathbf{A} - \mathbf{x}_i\mathbf{x}_i^T)^{-1}(\mathbf{X}^T\mathbf{y} - \mathbf{x}_i y_i) = \Big(1 + \frac{h_{ii}}{1 - h_{ii}}\Big)\mathbf{x}_i^T\mathbf{A}^{-1}(\mathbf{X}^T\mathbf{y} - \mathbf{x}_i y_i) \\ &= \frac{\hat{y}_i - h_{ii}y_i}{1 - h_{ii}}.\end{aligned}$$

Step 3. $y_i - \hat{y}_{(i)} = \dfrac{(1 - h_{ii})y_i - \hat{y}_i + h_{ii}y_i}{1 - h_{ii}} = \dfrac{y_i - \hat{y}_i}{1 - h_{ii}} = \dfrac{e_i}{1 - h_{ii}}$. ■